

Jam Genome — data report

Generated from `analysis/report.py`. Covers everything collected to **09 September 2026**.

What this dataset is

Segment speeds from the TomTom Traffic Flow Segment Data API, polled every 30 minutes across Mumbai. Each reading gives a road segment's `currentSpeed` and its `freeFlowSpeed`; the working measure throughout is

speed ratio = current speed ÷ free-flow speed, and a point counts as congested when that ratio is below **0.5**.

93,839 usable readings over 51 days. Mumbai has no open traffic dataset — Uber Movement shut down in 2023 — so this was collected from scratch.

Two eras, not one dataset

The monitoring points changed part-way through, so the two periods are reported separately and only compared where honest:

era	dates	points	days	readings
A	10 Jul 2026 – 10 Aug 2026	36	32	50,453
B	22 Aug 2026 – 09 Sep 2026	50	19	43,386

Era A used hand-picked junctions. Era B used points re-sited onto short segments after a citywide survey, for the reason set out next.

Headline result: most of the original points were blind

In era A, **only 19 of 36 points ever recorded congestion**. The other 17 sat at free-flow for a month — including Kalanagar, Amar Mahal and King's Circle, three of Mumbai's most notorious junctions.

They were not uncongested. Every TomTom reading describes a whole **road segment**, and Mumbai's expressways are modelled as single segments 13–20 km long. A jam at one junction is averaged away across the whole stretch. Segment length, not sampling rate, is the real resolution limit.

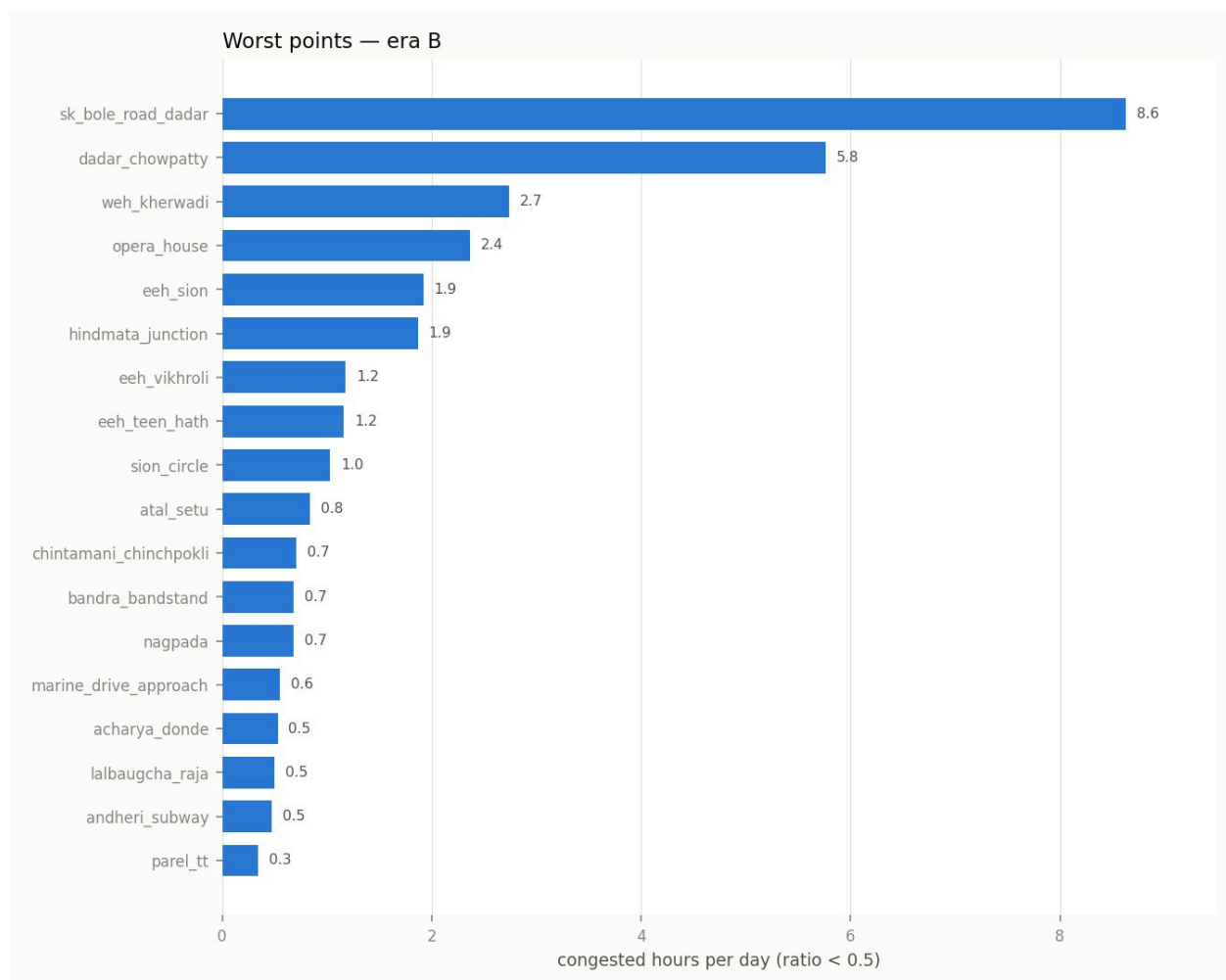
A 260 m-grid survey of Greater Mumbai (6,327 probes, `sweep/`) enumerated **870 distinct segments**, of which **692** are short enough (0.1–3 km) to measure a single junction. The points were re-sited onto those.

It worked: in era B, **42 of 50 points record congestion** — 84% versus 53% before.

Worst points

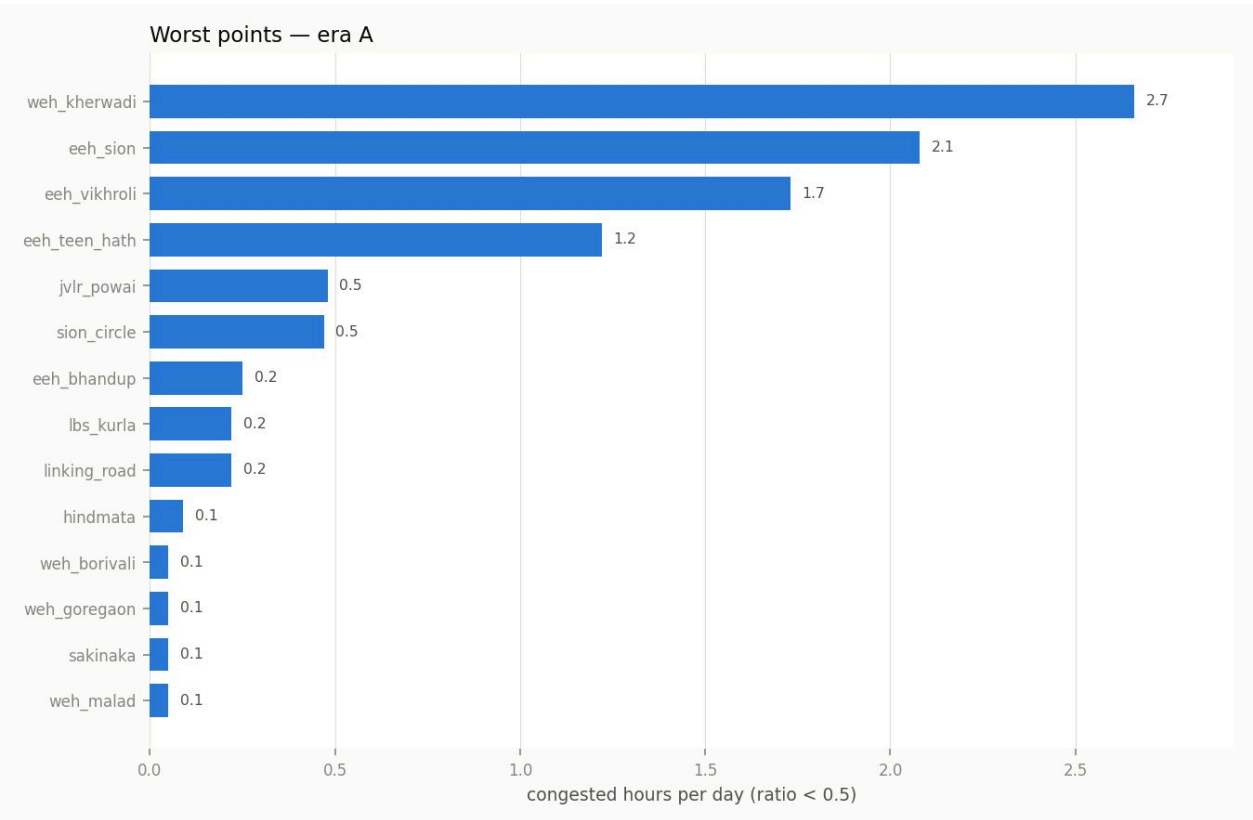
Era B (19 days, 50 points)

point	congested h/day	mean ratio	5th pct
sk_bole_road_dadar	8.63	0.67	0.23
dadar_chowpatty	5.76	0.74	0.30
weh_kherwadi	2.74	0.83	0.37
opera_house	2.37	0.79	0.42
eeh_sion	1.92	0.80	0.46
hindmata_junction	1.87	0.77	0.46
eeh_vikhroli	1.18	0.89	0.49
eeh_teen_hath	1.16	0.79	0.48
sion_circle	1.03	0.87	0.51
atal_setu	0.84	0.95	0.66
chintamani_chinchpokli	0.71	0.77	0.50
bandra_bandstand	0.68	0.87	0.53

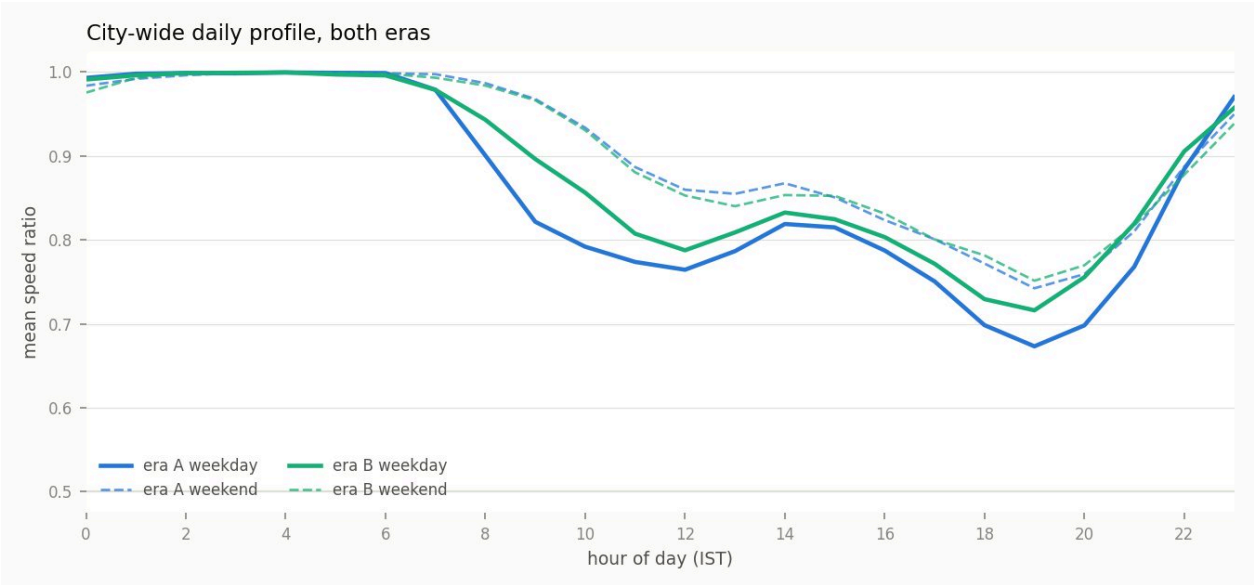


Era A (32 days, 36 points)

point	congested h/day	mean ratio	5th pct
weh_kherwadi	2.66	0.81	0.39
eeh_sion	2.08	0.79	0.46
eeh_vikhroli	1.73	0.88	0.44
eeh_teen_hath	1.22	0.79	0.47
jvlr_powai	0.48	0.88	0.56
sion_circle	0.47	0.89	0.57
eeh_bhandup	0.25	0.90	0.63
lbs_kurla	0.22	0.92	0.62
linking_road	0.22	0.82	0.55
hindmata	0.09	0.93	0.67



When Mumbai jams



The city shows a double hump — a late-morning dip and a deeper evening trough around 18:00–20:00 — with free-flow from roughly midnight to 07:00. Weekends are not uniformly better: in central Mumbai the weekend afternoon is frequently *worse* than the weekday commute.



Congestion episodes

An episode is two or more consecutive 30-minute slots below the threshold.

- era A: **134 episodes**
- era B: **215 episodes** over a shorter window

Longest single episodes in era B:

point	start (IST)	duration	worst ratio
sk_bole_road_dadar	22 Aug 12:00	12.0 h	0.15
sk_bole_road_dadar	25 Aug 11:00	11.0 h	0.26
sk_bole_road_dadar	01 Sep 11:00	11.0 h	0.15
sk_bole_road_dadar	29 Aug 12:00	10.5 h	0.15
sk_bole_road_dadar	06 Sep 12:00	10.0 h	0.09
sk_bole_road_dadar	02 Sep 13:00	9.0 h	0.11
acharya_donde	29 Aug 10:30	9.0 h	0.05
sk_bole_road_dadar	03 Sep 10:30	9.0 h	0.15

Threshold sensitivity

The 0.50 cut-off is a choice, so here is the whole curve rather than one number:

threshold	episodes	points affected	total hours
0.40	101	18	216
0.45	153	23	340
0.50	215	30	476
0.55	389	36	812
0.60	590	43	1322
0.65	850	45	2087
0.70	1042	47	2935

0.50 is selective enough to be meaningful and loose enough to leave signal; 0.40 is too sparse and 0.65+ flags nearly every point.

Deeper statistics (era B)

How long a jam lasts

Of the **215 jams** in era B: the median lasts **1.5 hours**, the average **2.2**, and the longest ran **12 hours**. **41%** are over within an hour, but **15%** drag on past three.

So most jams are short, but a meaningful tail runs for hours — and the worst single episode filled most of a working day.

Morning peak or evening peak?

48 of 50 points are worse in the evening than the morning. The most evening-skewed roads:

point	morning ratio	evening ratio
<code>dadar_chowpatty</code>	0.87	0.48
<code>sk_bole_road_dadar</code>	0.73	0.36
<code>sion_circle</code>	0.97	0.66
<code>eeh_vikhroli</code>	0.96	0.65
<code>ctl_jogeshwari</code>	0.93	0.63

Only 2 points suffer more in the morning, and they are unusual cases: `atal_setu` is the harbour bridge, which carries an outbound morning flow, and `eeh_bhandup` sits on a long expressway segment.

Which roads are commuter roads?

Comparing daytime (08:00–21:00) weekdays against weekends:

point	weekday	weekend	weekend is better by
<code>eeh_vikhroli</code>	0.78	0.93	+0.15
<code>weh_kherwadi</code>	0.68	0.82	+0.15
<code>jvlr_powai</code>	0.77	0.91	+0.14
<code>marine_drive_approach</code>	0.82	0.94	+0.12
<code>opera_house</code>	0.63	0.74	+0.11

13 of 50 points are actually worse at the weekend. Central Mumbai's shopping and leisure traffic outweighs the weekday commute on those roads — a genuine finding, and the opposite of the usual assumption.

The single worst hour of the week

day	hour	mean ratio
Tuesday	19:00	0.701
Friday	19:00	0.707
Monday	19:00	0.720
Wednesday	18:00	0.720
Tuesday	18:00	0.721

Every one of the ten worst hours falls between 18:00 and 19:00. **Tuesday evening is the worst hour in Mumbai's week.**

Which points jam together

Correlation of congestion between every pair of points. This shows **association, not causation** — with 30-minute sampling it cannot say which jam came first, which is the analysis the project never got the data density to complete.

correlation	point A	point B
+0.61	<code>dadar_chowpatty</code>	<code>sk_bole_road_dadar</code>
+0.39	<code>sk_bole_road_dadar</code>	<code>weh_kherwadi</code>
+0.30	<code>opera_house</code>	<code>sk_bole_road_dadar</code>
+0.29	<code>eeh_vikhroli</code>	<code>opera_house</code>
+0.29	<code>eeh_vikhroli</code>	<code>sk_bole_road_dadar</code>
+0.29	<code>dadar_chowpatty</code>	<code>eeh_teen_hath</code>

The strongest pair by a wide margin is `dadar_chowpatty` and `sk_bole_road_dadar` at **+0.61** — two roads about a kilometre apart in Dadar. That is the closest thing to propagation evidence in this dataset.

Did rain or events explain any of this?

Mumbai's monsoon ran through the whole collection window, and news reports for the period describe repeated disruption: over 400 mm in the first four days of July, a 205 mm day, a road cave-in on LBS Marg, and standstills on the Western Express Highway near Vile Parle on 18 August. The Sion road-over-bridge was also shut for reconstruction throughout (closed Aug 2024, due to reopen 30 Sept 2026), and the Sant Rohidas Marg subway opened mid-collection on 14 July 2026.

Rainfall was backfilled from the Open-Meteo ERA5 archive rather than the collector's own rain log, which is unusable (see Limitations). Testing at hourly resolution and **matching wet hours against dry hours of the same hour-of-day** — because rain and traffic both follow daily cycles — gives:

era	hours tested	wet hours	wet minus dry
A	716	307	-0.10 pp
B	436	65	-0.43 pp

No measurable increase in congestion on wet hours. If anything the sign is negative, and a naive pooled figure of -0.70 pp turned out to be mostly an artefact of mixing the two eras. Daily correlations are similarly flat (era A $r = -0.03$ over 29 days).

Three reasons not to read this as "rain doesn't cause jams":

1. Mumbai's heaviest rain days coincide with school and office closures, so demand falls at the same time capacity does.
2. ERA5 is a ~9 km grid and smooths the local cloudbursts that actually flood individual junctions.
3. Many points sit on long segments that average away a flooded stretch.

The honest statement is that **at this measurement resolution, rain has no detectable net effect on speed ratios**. Reproduce with `python analysis/rain_check.py`.

Limitations

1. **The two eras are not directly comparable.** Different point sets, and many era-A points measured segments too long to be informative.
2. **The rain log is unusable.** `collector/weather.py` sampled a 15-minute window every 30 minutes, so it missed roughly half of all rainfall. Any rain analysis must backfill from the Open-Meteo archive instead.
3. **No festival data.** The collection was designed around Ganeshotsav (14–25 Sept 2026), but TomTom began enforcing a 20,000 requests/month free-tier limit and the September quota was exhausted on 8 September. The festival window was not captured.
4. **Expressway propagation cannot be measured.** WEH and EEH are single segments, so every point on one returns the same value. Propagation analysis is only possible on the arterial and junction network.
5. **Ratios depend on TomTom's free-flow baseline**, which is their modelled figure, not a measured overnight speed.

Reproducing

```
python analysis/report.py          # this report and its figures
python analysis/eda.py             # league table, heatmaps, profiles
python analysis/episodes.py        # episode extraction + threshold sweep
python analysis/data_quality.py    # coverage and dead-point checks
```

Background and constraints are in [FINDINGS.md](#); the build plan is in [PLAN.md](#).